

Linear Algebra

in Two Pages

A demonstration document for the md4x STEM textbook template — typography, structure, and mathematical notation.

Anonymous

A demonstration document for the STEM textbook template — academic typography, numbered structure, theorem-style callouts, and inline mathematical notation.

0.1 Vectors

A vector $\vec{v} \in \mathbb{R}^n$ is an ordered n -tuple of real numbers. Vectors form the basic objects of linear algebra. They can be added componentwise and scaled by a real number, operations that obey the usual associative, commutative, and distributive laws.

0.1.1 Definition

Formally, a real vector space is a set V equipped with two operations: vector addition $+: V \times V \rightarrow V$ and scalar multiplication $\cdot: \mathbb{R} \times V \rightarrow V$, subject to eight axioms.

0.1.2 Operations

The inner product (or dot product) of two vectors $\vec{u}, \vec{v} \in \mathbb{R}^n$ is

$$\langle \vec{u}, \vec{v} \rangle = \sum_{i=1}^n u_i v_i.$$

The norm is $\|\vec{v}\| = \sqrt{\langle \vec{v}, \vec{v} \rangle}$. These two definitions give us the geometric notions of length and angle in $\mathbb{R}^n$.

Theorem 1.1 (Cauchy–Schwarz inequality). For any vectors $\vec{u}, \vec{v}$ in an inner product space,

$$|\langle \vec{u}, \vec{v} \rangle| \leq \|\vec{u}\| \cdot \|\vec{v}\|,$$

with equality if and only if $\vec{u}$ and $\vec{v}$ are linearly dependent.

The Cauchy–Schwarz inequality is a foundational tool — it underpins the triangle inequality, the convergence of series in normed spaces, and bounds in probability such as the covariance inequality.

0.2 Matrices

A matrix $A \in \mathbb{R}^{m \times n}$ is a rectangular array of real numbers with m rows and n columns. We write A_{ij} for the entry in the i -th row and j -th column.

0.2.1 Multiplication

Given $A \in \mathbb{R}^{m \times n}$ and $B \in \mathbb{R}^{n \times p}$, their product $AB \in \mathbb{R}^{m \times p}$ has entries

$$(AB)_{ij} = \sum_{k=1}^n A_{ik} B_{kj}.$$

Matrix multiplication is associative but not commutative: in general $AB \neq BA$.

0.2.2 A naive implementation

The textbook definition translates directly into nested loops. The following Python is correct but inefficient — production code should use a tuned BLAS routine.

```
import numpy as np

def matmul(A, B):
    m, n = A.shape
    n2, p = B.shape
    assert n == n2, "inner dimensions must match"
    C = np.zeros((m, p))
    for i in range(m):
        for j in range(p):
            for k in range(n):
                C[i, j] += A[i, k] * B[k, j]
    return C
```

0.2.3 Asymptotic cost

The naive algorithm above runs in $O(mnp)$ time. For square matrices ($m = n = p$) this is $O(n^3)$. Strassen's algorithm (1969) reduces the exponent to $\log_2 7 \approx 2.807$, and the current state of the art (as of 2024) achieves $O(n^{2.371552})$, though the constants make it impractical below astronomical sizes.

Algorithm	Year	Exponent
Naive	classical	3
Strassen	1969	2.807
Coppersmith–Winograd	1990	2.376
Williams et al.	2024	2.371552

0.3 Conclusion

This document demonstrates the STEM template's treatment of mathematical notation, numbered structure, theorem-style callouts, code listings, and tables. The body type is Charter; headings are Helvetica; math is rendered by KaTeX; and a single navy accent (#1e3a5f) carries the visual hierarchy.